



The influence of matrix type in the relationship between patch size and amphibian richness: A global Meta-Analysis

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ABSTRACT

The species-area relationship (SAR) is one of the main topics debated in ecology. Amphibians are highly sensitive to habitat loss and fragmentation, due to their traits and reproductive requirements. In the present study, we developed a meta-analysis to address a mean effect size to SAR regarding forest amphibians in man-made fragmented landscapes, and tested the influence of seven in-study and landscape variables that may affect this relationship globally. We conducted a search using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) protocol and calculated an effect size for each one of the 28 studies elected by our criteria. There is a positive relationship between amphibian richness and patch area, with a mean effect size of 0.68 and high heterogeneity among studies. The meta-regressions showed that only the matrix type surrounding the patches affected the variance between studies, explaining 46%. Clear cut was the matrix with the highest effect size, followed by silviculture, but both had a low representation of studies. Agriculture, grazing and urban matrices had a significant influence on patch area and the species richness relationship, but with distinct mean effects sizes and variances between-studies. The matrix type around the remnants may substantially affect results and interpretation of the SAR, possibly due to the varying capacity of amphibian species to cross and inhabit each matrix type. If possible, SAR should be compared within a context of multiple landscapes, and not only with a patch-focused and local approach.

1. Introduction

The species-area relationship (SAR) was described in the first years of 20th century (Arrhenius, 1921) and since then has been one of the main topics in ecology (Preston, 1948; Darlington, 1957). The SAR suggests that communities with reduced area will have reduced species richness. This relationship has been screened and tested extensively in the field of conservation biology with forest patches surrounded by a human-made matrix (MacArthur and Wilson, 1967; Diamond, 1975; Wilson and Willis, 1975; Zimmerman and Bierregaard, 1986). Despite the non-natural surroundings, these patches are not completely isolated from each other as in an impenetrable matrix (Laurance, 2008). The matrix surrounding patches has a degree of permeability to certain species, and can therefore directly affect species composition inside

patches (Ricketts, 2001; Prugh et al., 2008; Dixo and Metzger, 2010; Koh and Ghazoul, 2010; Lindsay et al., 2013; Thompson et al., 2016). Many other aspects interfere in the species composition inside the forest remnants: geographical location (e.g., peninsula effect), time since the fragmentation event, local disturbance history (e.g., fire patterns), environmental synergism, remnant shape, habitat heterogeneity inside the patches, species traits, presence of roads and even climate (Boecklen and Gotelli, 1984; Swetnam et al., 1999; Lindell et al., 2007; Laurance, 2010; Rytwinski and Fahrig, 2012; Battisti, 2014; Keinath et al., 2017).

Watling and Donnelly (2006) quantified several aspects of the relationships between species richness and area together with isolation in a meta-analysis regarding multi-taxa populations inside patches of natural habitat (not restricted to forest remnants). In their review, 118 studies tested the SAR, where 91% of studies showed a significant and

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positive influence of area in species richness, 1% had a significant and negative influence and 8% had no significant results. They found that area explained more variance in species richness than isolation, and that it was more strongly correlated with species differences between patches. In contrast, another meta-analysis by Prugh et al. (2008) reported that area and isolation were, surprisingly, poor predictors of species occupancy when analyzed individually or combined (global meta-analysis, $n = 55\,855$ occupancy records). Area predictors of species occupation were strongly influenced by matrix type, and could vary according to the species' taxonomic group and traits (Prugh et al., 2008).

Amphibians are the most threatened vertebrate species worldwide, with 31.5% of species categorized as vulnerable, endangered or critically endangered (IUCN, 2019). In comparison with others vertebrate taxa, amphibians are impacted more negatively by habitat loss and fragmentation due to their relatively low movement capacity, physiological traits and habitat requirements (Stuart et al., 2004; Cushman, 2006). The majority of amphibians species have a larval stage, which makes them dependent on water bodies (e.g. ponds, streams, tank bromeliads) and highly sensitive to habitat fragmentation (Becker et al., 2007). Amphibian species with direct development (for example those from the so called group Terrarana, with 989 species) are not directly dependent on water bodies. They do however need a minimum environmental humidity due to their sensitive skin and gelatinous eggs, a requirement that a forested environment fulfills (Vitt and Caldwell, 2013; Watling and Braga, 2015). Terrestrially developing species are much more sensitive to habitat loss than habitat disruption (Becker et al., 2007). A vote-counting study conducted by Almeida-Gomes et al. (2016b) found that 11 out of 14 studies (considering only the studies which tested the SAR for amphibians) showed a significant positive effect of the forest patch area on species richness, with just a few studies discovering a null effect (e.g., Hillers et al., 2008) and none reporting a negative effect. For amphibians inhabiting wetlands, a meta-analysis performed by Quesnelle et al. (2015) demonstrated that the amount of surrounding forest was a more important predictor for species abundance than the amount of wetland itself, or the matrix composition (d.f. = 204). Habitat loss and fragmentation lead to reduce amphibian reproduction, due to the reduction in the availability of reproductive habitats (Zimmerman and Bierregaard, 1986; Becker et al., 2007; Almeida-Gomes and Rocha, 2015; Almeida-Gomes et al., 2016b).

Habitat loss is one of the greatest threats to amphibian species from forested habitats (Stuart et al., 2004; Cushman, 2006; Van Sluys et al., 2009; Pfeifer et al., 2017). With the constant reduction of forest habitats and the isolation of populations into small remnants (Haddad et al., 2015), we need a deeper understanding of the mechanisms behind the SAR in forest-sensitive species. In the present study, we developed a meta-analysis to (1) determine the mean effect size to SAR regarding amphibians from forested areas within man-made fragmented landscapes worldwide, and (2) test in-study and landscapes variables that may affect this relationship globally.

2. Material and methods

2.1. Literature search and eligibility criteria

We recorded the number of studies identified and the number of studies included or excluded in each step according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement (PRISMA Statement; Moher et al., 2009). We searched the databases ISI Web of Science and Scopus until November 2018, with a combination of keywords (*amphibi** OR *anur**) AND (*fragmentation* OR *patch size*), and no year restriction ("identification" step). Despite having used English keywords, we did not exclude studies from other languages if the statistical reports were intelligible. We also searched in the cited literature section of the studies captured in our search to refine it and find relevant papers not captured in the original database search. Each

study had its title and abstract screened ("screening" step). The study proceeded into reading of the full text when we understood that it was related to habitat loss and/or fragmentation and amphibian communities. Studies were not included in the analyses if they were not on amphibians or habitat loss.

Next, we applied our eligibility criteria ("eligibility" step, as follows). We excluded the following types of studies: data from naturally fragmented areas (e.g., natural islands formed as a consequence of Pleistocene's climate changes), data from non-forested habitats (e.g., South America grassy plains "Pampas"), data from reforested patches or patches with selective cutting, systematic reviews, meta-analysis, computer models and simulations. Despite the exclusion, we screened their "cited literature" section as well. If two or more studies had a common data set (e.g., Watling and Donnelly, 2008; Watling et al., 2009), we only considered the study with the highest number of patches, or the one with the necessary data available to extract the correlation coefficient. As we focused our search on the relationship between patch size and amphibian species richness, we did not use data from studies that measured occupancy, means and variance between groups (e.g., richness in forested habitat vs. non-forested habitats). We included articles which studied the amphibian communities inside forest remnants, as long as they fulfilled the previous eligibility criteria and the authors made the data available as a result, either as supplementary material or by direct contact with the corresponding author. We used this available information to calculate the correlation coefficient with correlation analysis (explanation below).

2.2. Data extraction and meta-analysis

We calculated an effect size for each study eligible for meta-analysis. As our question is based on the relationship between two continuous variables (patch size and amphibian richness), we calculated the effect size based on correlation coefficients from correlation and regression analysis (Borenstein et al., 2009; Rosenberg et al., 2013). To calculate the effect size and its variance between studies, we extracted the correlation coefficient and the degree of freedom from the respective analyses. We extracted the Pearson's r or the Spearman's ρ values by direct citation in the text, tables or figures, or by removing the square root from the R^2 value. We did not use slope coefficients, AIC values or partial R^2 values from multiple regression in our meta-analysis. Instead, we accessed the raw available data and calculated the Pearson's r or the Spearman's ρ (description below). Then, we converted all the r and ρ values to Fischer's z -scale, in order to normalize the correlation coefficient distribution and have an independent variance between studies (Borenstein et al., 2009; Rosenberg et al., 2013). We calculated the effect size and reported the present meta-analysis in Fischer's z -scale.

Some of the selected papers did not test the relationship between patch size and amphibian species richness, or did a correlation/regression analysis but reported neither the r/ρ nor the R^2 value. If the raw data was available, we included these studies in the meta-analysis by calculating the Pearson's r or Spearman's ρ . If the raw data was not available, we contacted the corresponding author asking for the raw data. We first tested the data for normality, homoscedasticity and linearity, and if they fulfilled all those prerequisites, performed a Pearson correlation analysis between a log-transformed patch size and amphibian richness. If these prerequisites were not fulfilled, we performed a Spearman ranking correlation (non-parametric) instead. We did not consider studies with less than five patches, because it was not possible to estimate sampling variance with a correlation for four or less samples (Viechtbauer, 2010).

We applied a random-effect (RE) model to calculate the general effect size among all the studies and to explore possible sources of heterogeneity between studies. We opted to proceed with a random-effect model instead of fixed-effect model because: 1 - The fixed-effect model assumes that the true effect size is the same between all studies

and the sampling error is zero (Borenstein et al., 2009; Gurevitch et al., 2018), which we cannot assume in the present meta-analysis; 2 - heterogeneity is expected between ecological studies (Senior et al., 2016; Gurevitch et al., 2018); and 3 - the studies were not homogenous between them (Q test for heterogeneity: $Q = 93.22$, d.f. = 27, $p < 0.0001$). Such heterogeneity was inspected with meta-regressions (Borenstein et al., 2009; Rosenberg, 2013). We tested the influence of moderators in the variance between studies with regression models (Borenstein et al., 2009; Rosenberg, 2013). We applied individual meta-regression models for each moderator and a global RE model, which includes all the moderators. If a categorical moderator was significant, we also applied individual RE models for each category as a method to track and understand part of the between-studies variance (Borenstein et al., 2009; Rosenberg, 2013; Gurevitch et al., 2018).

We use Cochran's Q statistics (Cochran, 1954; Borenstein et al., 2009) to evaluate and test the variance among k studies. The Q value represents the observed variation to the within-study error. The τ^2 and τ values are the between-studies variance and its standard definition, respectively. The I^2 is the proportion of the observed variance between studies, accessed by the formula $[(Q - d.f.)/Q] * 100$. The I^2 ranges from 0 to 100%, measuring the degree of inconsistency across studies, where higher levels of heterogeneity are represented by a high I^2 value (Higgins and Thompson, 2002). In meta-regression models, the total amount of heterogeneity (Q_T , from Q_{Total}) can be partitioned into heterogeneity from the structural model Q_M (from Q_{Model}) and the unexplained heterogeneity Q_E (from Q_{Error}), where $Q_T = Q_M + Q_E$. In simple models (without moderators), there is no explanatory variable, so $Q_M = 0$ and $Q_T = Q_E$. To test the presence of heterogeneity between studies, we used Q tests for heterogeneity, which consist in a chi-distribution with $k-1$ degrees of freedom and tested the hypothesis of homogeneity between studies (Cochran, 1954; Borenstein et al., 2009). A significant Q test means heterogeneity in the variance between-studies. In the meta-regression model summaries, the R^2 value represents the amount of I^2 explained by the model (Borenstein et al., 2009).

We tested the influence of seven moderators in the relationship between patch size and amphibian species richness: Number of patches considered in correlations, patch size range, survey method, latitude, global biome (*sensu* Lomolino et al., 2016), the total amphibian species richness in the respective landscape and dominant surrounding matrix (Table 1). To evaluate the possible influence of the survey method, we categorized the methods based on the nature of the search for amphibians. We defined any kind of active detection for amphibians by the researchers, such as visual and auditory encounter surveys, transects and plots, as "active" (Heyer et al., 1994); any kind of detection based in fixed points, such as pitfall traps, net trapping and audio recorders, as "passive"; and studies that used both types of methods without distinguished results (e.g. visual encounter survey combined with pitfall

traps) as "active-and-passive". We tested the possible influence of the landscape distance from Equator line with the latitude coordinates given by the studies. If the coordinate was not provided in the text, we used the median latitudinal coordinate from the described municipality or region. We converted the given coordinates to decimal in WGS84 datum, and converted negative values to positive. We assumed the dominant surrounding matrix as the predominant land use as defined by the authors. If the study did not specify the land use in the surroundings, we consulted other contemporary studies that were performed in the same locality and/or local Governments official websites. We categorized the land use similarly to Thompson et al. (2016), as described in Table 1. We could not appoint the time of fragmentation as a moderator because few eligible studies could address the precise moment of fragmentation in their respective landscape.

We tested publication bias with Rosenberg, Rosenthal and Orwin fail-safe N tests (Rosenthal, 1979; Orwin, 1983; Rosenberg, 2005) and regression tests for funnel plot asymmetry (Egger et al., 1997) for each RE model. We considered a test as significant when $p < 0.05$. All the analyses were made in R version 3.5.2 (R Core Team, 2019). The correlation analyses were made with the *cor* function (*stats* package; R Core Team, 2018) and the meta-analysis were made with functions from the *metafor* package (Viechtbauer, 2010).

3. Results

We found a total of 1333 matches (excluding duplicates) with the selected key-words, but only 441 studies proceeded after the screening step (Fig. 1). After the eligibility and inclusion steps, 28 studies were selected for the meta-analysis (Table 2). We collected the correlation coefficient directly from 11 studies, and calculated it independently from the others 17. These 28 studies were carried out in 13 different countries: South (8) and North (5) America, Asia (5), Africa (4), Central America (2), Australia (2) and Europe (2).

3.1. Effect size and publication bias

There was a highly significant positive overall influence of patch size on amphibian species richness, with a mean effect size of 0.68 (Random-effect model; $k = 28$; 95% CI: 0.48–0.88; $p < 0.0001$). In fact, 26 (92.8%) of the effect sizes were positive and only two (7.2%) were negative (Fig. 2). However, there was significantly high heterogeneity among studies, with an I^2 value over 70% (RE model, Table 3). Despite the majority of landscapes coming from the American continent (15) and from tropical rainforests (17), we did not detect publication bias among studies (Table A for the N-fail safe tests), with a symmetrical distribution of the effect sizes around the mean effect size (Test for funnel plot asymmetry: $z = 1.23$, $p = 0.22$; Figure B and Table B).

Table 1
Variables moderators insert in the present meta-analysis.

Moderator variable	Variable type	Variable description
Number of patches	Continuous	Total number of patches considered in the regression analysis.
Patch area range	Continuous	The difference between the largest and the smallest patch.
Sample method	Categorical	Survey method used to detect species: <i>Active</i> - Visual and/or auditory encounter surveys, transects, plots, any survey based in active search for individuals. <i>Passive</i> - Pitfall traps, net trapping, audio recorders, any survey based in a fixed point. <i>Active-and-passive</i> - The use of both methods (e.g., visual encounter surveys and pitfall).
Latitude	Continuous	Distance from the Equator (decimal, datum WGS84).
Biome	Categorical	Biome zone according to Lomolino et al. (2016).
Total amphibian Richness	Continuous	Total number of species recorded in the landscape.
Matrix type	Categorical	Dominant matrix surrounding the patches, according to the study's authors' description: <i>Agriculture</i> - Crop plantations; <i>Clear cut</i> - Clear cut followed by abandonment; <i>Flooding</i> - Patches as artificial islands, as a direct consequence of artificial dams; <i>Grazing</i> - Pasture land for animal farm; <i>Silviculture</i> - Cacao, shadow coffee, <i>Eucalyptus</i> , <i>Pinus</i> and/or rubber plantations; <i>Urban</i> - Land occupation for human residential use.

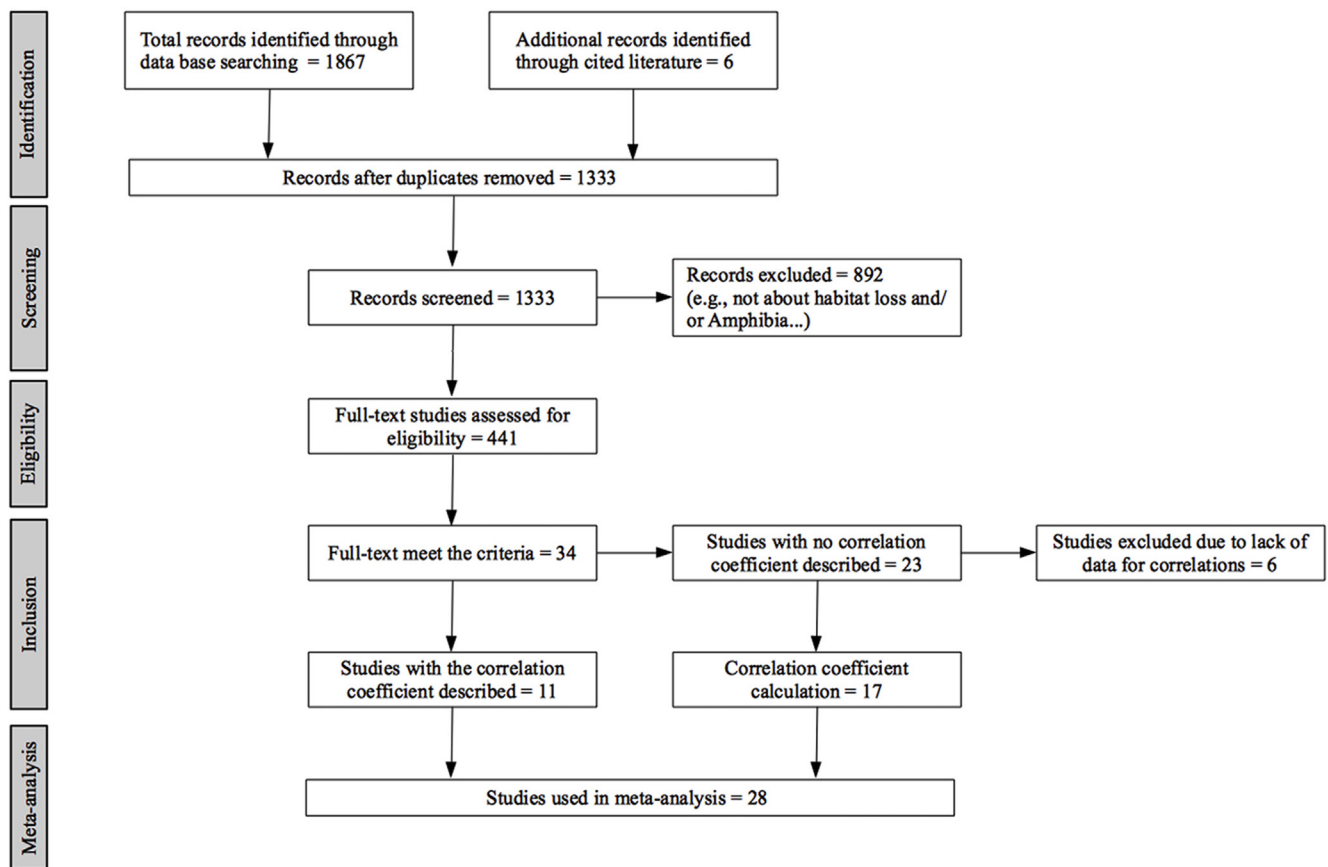


Fig. 1. Flow diagram indicating the number of studies considered and excluded at each step, as suggested by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement (PRISMA Statement). Each gray box indicates the respective step.

3.2. Meta-regressions

The individual meta-regression analysis indicates that only the matrix type surrounding the patches could explain part of the variance between studies (Table 3). By adding the matrix type as a moderator, the I^2 drops from 71% to 56%. Also, matrix type could explain 45.9% of the inconsistency variance between studies. All other moderators explained less than 0.01% of the variance between studies when tested in individual models, with I^2 values above 70%. A global model reduces the I^2 to 62.2% and explains 13.1% of the inconsistent variance.

The influence of matrix type on the between-studies variance is notable when we aggregated each study according to matrix type (Fig. 2). The most representative matrix type was agriculture (nine, 32.1%), followed by urban (eight, 28.6%) and grazing (six, 21.4%). After partition, the model representing only studies with agriculture or grazing matrices reduces its I^2 to below 50% (Table 4). Clear-cut was the matrix with the highest effect size (effect size = 2.65, 95% CI: 1.65, 3.63; Figs. 2 and 3) followed by silviculture (mean effect size = 1.01, 95% CI: 0.27, 1.76). The lowest mean effect size was found in the agriculture-only model (= 0.24, 95% CI: 0.01, 0.47), followed by the grazing-only model (= 0.78, 95% CI: 0.41, 1.15). With the exception of the flooding-only model, the models' confidence intervals did not cross the null effect line (when Fischer's z-value is equal to 0; Figs. 2 and 3), indicating that the significant positive relationship between patch size and amphibian richness was maintained, but with a distinct degree of influence according to matrix type (Table 4). We did not detect publication bias when we separated the collection of studies according to the matrix type (Figure B and Table B; we could not evaluate publication bias in the categories with less than three studies).

4. Discussion

A meta-analysis as used in the present study has two main objectives: to detect and describe ecological patterns pointing out the central tendency and the heterogeneity of mean effect across studies (Harrison, 2011; Gurevitch et al., 2018). The interpretation of the current meta-analysis in our study indicated a significative and positive relationship between forest patch size and richness of amphibians. We detected the heterogeneity in the selected set of data and found that matrix type was the main source of such heterogeneity among the tested moderators.

Previous similar meta-analyses explored the habitat amount with amphibian species richness and abundance (e.g., Bender et al., 1998; Watling and Donnelly, 2006; Quesnelle et al., 2015). Our study focused on the association between forest amphibians and forest habitat availability. The biological response of species to habitat area varies according to the taxa, where species traits played a major role in the species' capacity to successfully access and occupy the patch (Bender et al., 1998; Watling and Donnelly, 2006; Prugh et al., 2008; Becker et al., 2010; Keinath et al., 2017). Forest is a major variable in a landscape for amphibians, even when they use and reproduce in habitats outside the forest patches (Quesnelle et al., 2015; Almeida-Gomes et al., 2016a).

4.1. Patch size influences amphibia richness

The positive relationship between habitat area and species richness was conceived and established before the Island Biogeography theory itself (Arrhenius, 1921; Darlington, 1957; MacArthur and Wilson, 1967). It also came to be supported recently by field and meta-analytical studies regarding habitat area (e.g., Watling and Donnelly, 2006; Prugh et al., 2008). The development of studies investigating the SAR

Table 2
Summary of studies used in meta-analysis. Studies with a “*” had their correlation coefficients (r , ρ or $\sqrt{R^2}$) directly extracted.

Study	Number of patches	Patch range (ha)	Matrix type	Total amphibian richness	Latitude	Biome	Method	Country
Li et al. (2018)*	16	170 (8–178)	Urban	6	31.2222	Temperate subtropical	active	China
Wang et al. (2018)	23	1288.61 (0.59–1289.2)	Flooding	9	29.6055	Temperate subtropical	active	China
da Silva and Rossa-Feres (2017)*	18	3308 (59–3367)	Agriculture	36	19.5666	Tropical rainforest	active-and-passive	Brazil
Dayananda et al. (2017)	18	13871.16 (1.71–13 872)	Silviculture	25	21.9166	Tropical rainforest	active	China
Ferrante et al. (2017)	16	208.74 (5.14–213.88)	Agriculture	21	–21.4256	Tropical rainforest	active	Brazil
Ferreira et al. (2017)	11	168.4 (1.8–170.2)	Urban	17	–20.4427	Tropical savanna	active-and-passive	Brazil
Almeida-Gomes et al. (2016b)	21	617.15 (1.87–619.02)	Grazing	50	–22.4	Tropical rainforest	active	Brazil
MacGregor-Fors et al. (2016)	6	92.3 (3.2–95.5)	Urban	10	19.5312	Tropical rainforest	active	Mexico
Russildi et al. (2016)	9	89 (3–92)	Agriculture	22	16.35	Temperate subtropical	active	Mexico
Lima et al. (2015)*	10	2137 (3–2140)	Flooding	32	–3.8316	Tropical rainforest	active-and-passive	Brazil
Riemann et al. (2015)	9	14.5 (2–16.5)	Agriculture	40	–21.2166	Tropical rainforest	active	Madagascar
Lion et al. (2014)*	23	25.7 (1.7–27.4)	Agriculture	14	–6.1089	Tropical rainforest	passive	Brazil
Cabrera-Guzmán & Reynoso (2012)	6	16 (1.4–17.4)	Grazing	10	18.6	Temperate subtropical	active	Mexico
Hilje & Mitchell Aide (2012)	15	10195.2 (2.4–10197.6)	Agriculture	29	10.4333	Tropical rainforest	active	Costa Rica
Bickford et al. (2010)	12	924 (11–935)	Urban	20	1.3516	Tropical rainforest	active	Singapore
Vignoli et al. (2009)*	62	1599 (1–1600)	Urban	15	41.8919	Temperate subtropical	active	Italy
Hilliers et al. (2008)*	16	46.5 (1.5–48)	Agriculture	19	5.75	Tropical rainforest	active	Ivory Coast
Watling & Donnelly (2008)	24	7.82 (0.64–8.46)	Grazing	13	–14.75	Tropical savanna	passive	Bolivia
Rakotoniravony (2007)*	11	4103 (47–4150)	Grazing	15	–13.2011	Tropical rainforest	active-and-passive	Madagascar
Bell and Donnelly (2006)*	9	5.45 (1.43–6.88)	Grazing	41	10.4333	Tropical rainforest	active	Costa Rica
Drinnan (2005)*	19	171.36 (0.24–171.6)	Urban	8	–34.05	Temperate subtropical	active	Australia
Alcala et al. (2004)	10	117 (5–122)	Agriculture	11	10	Tropical rainforest	active	Philippines
Pineda & Halfiter (2004)	10	111 (11–122)	Silviculture	21	19.5666	Tropical rainforest	active	Mexico
Silva et al. (2003)	11	39.7 (3.1–42.8)	Agriculture	9	46.4166	Tundra/Boreal/Alpine	active-and-passive	Canada
How & Dell (2000)	34	336.90 (1.1–338)	Urban	15	–31.9522	Temperate subtropical	passive	Australia
Vallan (2000)*	7	135.8 (0.16–136)	Grazing	25	–18.161	Tropical rainforest	active	Madagascar
Dickman (1987)	8	7.17 (0.23–7.40)	Urban	5	51.752	Temperate subtropical	active	England
Zimmerman and Bierregaard (1986)*	7	499 (1–500)	Clear cut	32	–3.1071	Tropical rainforest	active	Brazil

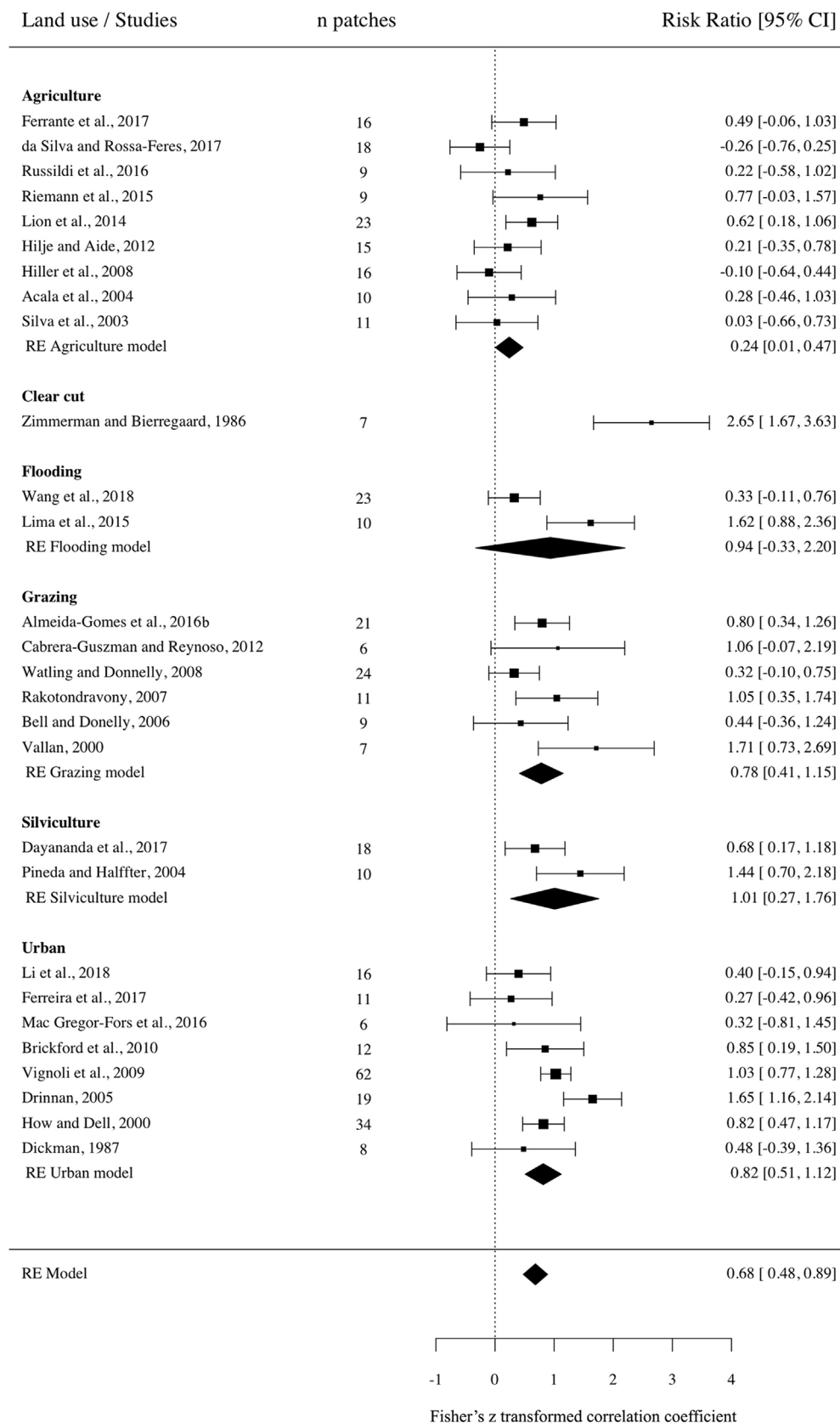


Fig. 2. Forest plot representing the individual effect sizes from each study (squares) and the mean effect size (diamond shape) from each Random Effect (RE) model. Error bars represent the 95% confidence interval. The dashed line represents the null effect.

Table 3

Test for residual heterogeneity and test of moderators among the random effect (RE) models. $\tau^2 \pm \tau$ = between-studies variance $\pm$ its standard definition; Q_E = unexplained heterogeneity; I^2 = the proportion of the observed variance between-studies. Q_M = moderator heterogeneity. In a model without moderators (RE model, first line), Q_E represents the total heterogeneity (Q_T). R^2 = the amount of I^2 explained by the model. The “*” means a significant p value.

Moderator	Test for residual heterogeneity				Test of moderators				
	$\tau^2 \pm \tau$	Q_E	d.f.	p value	I^2 (%)	Q_M	d.f.	p value	R^2 (%)
No moderators	0.1979 $\pm$ 0.0873	93.22	27	< 0.0001*	71.04	–	–	–	–
All variables	0.1719 $\pm$ 0.1085	37.07	14	0.0007*	62.24	21.26	13	0.0680	13.10
Number of patches	0.2121 $\pm$ 0.0911	87.50	26	< 0.0001*	70.29	< 0.01	1	0.9466	< 0.01
Patch size range	0.2065 $\pm$ 0.0926	92.15	26	< 0.0001*	71.78	0.24	1	0.6257	< 0.01
Sample method	0.2099 $\pm$ 0.0953	87.55	25	< 0.0001*	71.44	0.88	2	0.6427	< 0.01
Latitude	0.2204 $\pm$ 0.4695	93.03	26	< 0.0001*	72.05	1.19	1	0.2751	< 0.01
Biome	0.1996 $\pm$ 0.0928	80.77	24	< 0.0001*	70.29	2.65	3	0.4479	< 0.01
Total amphibia Richness	0.2104 $\pm$ 0.0937	93.14	26	< 0.0001*	72.08	0.26	1	0.6106	< 0.01
Matrix type	0.1070 $\pm$ 0.0623	49.95	22	0.0006*	55.95	21.48	5	0.0007*	45.90

has resulted in three main current hypotheses that explain the relationship between species richness and patch area. (1) The population hypothesis suggests that species can maintain larger populations in larger areas, and are less likely to become locally extinct in comparison to small populations from smaller areas (Shaffer, 1981). (2) The passive sampling hypothesis describes this relationship as a function of probability: a bigger patch in a fragmented landscape has a higher chance to be reached by individuals through dispersion (Coleman et al., 1982). (3) Alternatively, the habitat heterogeneity hypothesis proposes that larger areas have larger species richness due to higher availability of variable types of habitats (Boecklen and Gotelli, 1984; Ricklefs and Lovette, 1999; Tews et al., 2004). The habitat heterogeneity hypothesis is especially reasonable for amphibians, once their species richness had already been related to habitat heterogeneity, possibly due to the variety of reproductive modes (Keller et al., 2009; Vasconcelos et al., 2009; Bickford et al., 2010; Becker et al., 2010; Souza et al., 2014; Almeida-Gomes et al., 2016b; Araújo et al., 2018; Li et al., 2018; Figueiredo et al., 2019). Forest-specialist species (stream-dwelling, directly-development leaf-litter and some bromeligenous species) had a positive response to forest patch size, probably because there are more reproductive habitats available inside larger patches (Bender et al., 1998; Becker et al., 2007; Almeida-Gomes et al., 2014, 2016b; Almeida-Gomes et al., 2016a). On the other hand, generalist species can be more strongly associated with the amount of remaining edge (Bender et al., 1998; Ferrante et al., 2017; Ferreira et al., 2017; Figueiredo et al., 2019). The influence of matrix type only contributes to this discussion, because amphibian species' capacity to disperse across a human-made matrix is also related to each particular species-specific traits, such as body size, dispersal capacity, desiccation tolerance and habitat use (Fonseca et al., 2013; Watling and Braga, 2015; Thompson et al., 2016; Nowakowski et al., 2017).

4.2. (Trying to) interpreting variance between-studies

Ecological studies inherently have a high amount of variability among them (Senior et al., 2016), which is one of the main criticisms of

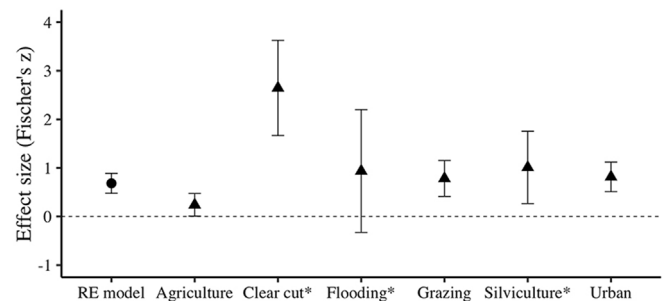


Fig. 3. Mean effect size for random effect models from each matrix type (triangles). RE model (circle) represents the complete random effect model, with all the studies. The dashed line represents the null effect. The error bars represent the 95% confidence interval. The “*” means a matrix type with less than three studies.

the use of meta-analysis in this field (Whittaker, 2010). Ecological and evolutionary meta-analytic studies had a mean I^2 between 83% and 92%, and less than 5% had an $I^2 = 0$ (Senior et al., 2016). However, detecting, labeling and explaining such sources of heterogeneity is one of the central questions when applying a meta-analysis to ecological and evolutionary studies (Senior et al., 2016; Gurevitch et al., 2018). In the present study, the I^2 was 71% in the RE model, which drops to 55% when we added the matrix type as a moderator. Close to 46% (R^2) of the 55.9% model heterogeneity (I^2) was explained by this moderator. This means that 25.7% of the total variance could be explained just by the matrix in the surrounding remnants. Others variables not explored in the present analysis could be important moderators to explain additional portions of the heterogeneity, such as isolation, time since fragmentation and availability of water bodies inside the remanent (Watling and Donnelly, 2006; Prugh et al., 2008; Bickford et al., 2010; Chen and Peng, 2017; Semlitsch, Walls, Barichivich, & O'Donnell, 2017). We could not appoint them as a moderator in the present study due to the lack of information and the complexity of each topic. For example, only the flooding matrix studies (Lima et al., 2015; Wang

Table 4

Individual random-effect models for each matrix type. $\tau^2 \pm \tau$ = between-studies variance $\pm$ standard definition; Q_T = the total amount of heterogeneity; k = total amount of studies considered; I^2 = the proportion of the observed variance between studies; z -value = Fischer's z value. The “*” means a significant p value.

Matrix type	Test for heterogeneity				Model summary			
	$\tau^2 \pm \tau$	Q_T	k	p -value	I^2 (%)	Estimate $\pm$ SE	z -value	p -value
Agriculture	0.0338 $\pm$ 0.0632	10.94	9	0.2049	26.90	0.24 $\pm$ 0.12	2.02	0.0433*
Clear cut	–	0	1	1	–	2.65	5.29	< 0.0001*
Flooding	0.7382 $\pm$ 1.1803	8.65	2	0.0033	88.45	0.94 $\pm$ 0.64	1.45	0.1467
Grazing	0.0891 $\pm$ 0.1315	9.02	6	0.1081	44.59	0.78 $\pm$ 0.19	4.15	< 0.0001*
Silviculture	0.1889 $\pm$ 0.4153	2.80	2	0.0941	64.32	1.01 $\pm$ 0.38	2.66	0.0078*
Urban	0.1057 $\pm$ 0.1026	18.52	8	0.0098	62.21	0.82 $\pm$ 0.15	5.24	< 0.0001*

Table 5

A summary of the intrinsic and local moderators and why they would influence species-area relationship (SAR) for amphibians in forest patches in the present meta-analysis.

Moderator	Nature	Explanation	Influence on SAR
Number of patches	Intrinsic to the study	A lower number of sampling units would result in a weak response to SAR (Almeida-Gomes et al., 2016b).	No
Patch area range	Intrinsic to the study	A large range of fragment sizes would result in a more robust response in the SAR (Almeida-Gomes et al., 2016b).	No
Sample method	Intrinsic to the study	Sample method would survey only a group of amphibians (e.g., pitfall traps surveys mostly leaf-litter amphibians; e.g., Siqueira et al., 2011), affecting the number of species detected in the patches (Almeida-Gomes et al., 2016b).	No
Latitude	Local variable	Regional species richness increases from the poles to the equator, so the SAR effect size should show a latitudinal decline (Hillebrand, 2004; Drakare et al., 2006).	No
Biome	Local variable	There is a sample bias, where amphibians from temperate forest were more surveyed than tropical and subtropical forests (Gardner et al., 2007); and/or The biodiversity of the Amphibia taxon is globally determined by the mean temperature, rainfall and its periodicity, where tropical forest holds most of the amphibia species (Vitt and Caldwell, 2013).	No
Total amphibian richness	Local variable	A lower local richness would reduce the variance in the correlation, making harder to detect a correlation if exist.	No
Matrix type	Intrinsic to the study	Species' tolerance to cross a human-made matrix and reach the forest patch is related to particular species-specific traits (James I. Watling and Braga, 2015; Thompson et al., 2016; Nowakowski et al., 2017).	Yes

et al., 2018) could address the exact moment of the fragmentation event, and few others tested the influence of water bodies inside the habitat remnants (e.g., Bickford et al., 2010; Li et al., 2018). We believe that isolation and extinction debt are complex topics that deserve their own reviews and meta-analyses, with a literature search and screening focused in exploring these points.

The low explanation from in-study and local variables (with exception of matrix, Table 3), indicates a possible global pattern from the SAR in amphibians in forested habitats. Assembling all of the variables in a global model explained 33% less than a model with only the matrix variable. We summarize which moderators would have influence on SAR in the present meta-analysis in a final table (Table 5). Almeida-Gomes et al. (2016b) suggested a possible influence of the sampling method (e.g., pitfall or plots) and sample design on the range of responses between previous studies, some of which are considered in the present meta-analysis (Hillers et al., 2008; Watling and Donnelly, 2008; Lion et al., 2014). The limited number of sampling units in fact provide a weaker response of amphibian richness to patch size in some studies (a small number of patches resulted in bigger errors in effect sizes; Fig. 2). Also, different sampling methods may result in obtaining distinct species compositions, a possibility that we did not explore in the present meta-analysis. However, neither method nor sample size influenced the meta-regression models (Table 3), indicating a lack of influence in the SAR regarding amphibians in forest habitats. Drakare et al. (2006) found a significant influence of latitude to SAR, with a reduction of z values from tropical to temperate regions. However, this tendency may vary according to the taxa as well, since we did not find this influence in our analysis. We did not detect publication bias among studies (Appencies A, B and C), so it is unlikely that this null result is due to the majority of studies being placed in tropical or sub-tropical regions. Our understanding of this collection of results is that the landscape had a greater influence on the SAR regarding amphibians in forested habitats than methodological and geographical aspects of each study.

4.3. Matrix is an important factor to consider in SAR studies

Human altered habitats are in a state of constant modification: At the moment of deforestation, the very first matrix type is probably a clear-cut, and may be followed by agriculture, grazing or urban matrices (Foley et al., 2005). After a long time of agriculture, exhausted soils may be converted for grazing or to urban matrix, or simply be abandoned. Our results only represent a moment of the patch lifetime since fragmentation, but elucidate part of this complex relationship

between area and species occurrence.

Increasing human activity results in a large conversion of original habitat to human-made landscapes, and among all the progress there are pieces of forest remaining. Amphibian individuals can be negatively affected by human interference at the landscape and local scales. At the landscape scale, the species need to cross different kinds of environments in order to access a habitat remnant. This usually expose them to unfavorable conditions such as desiccation and/or predation (Becker et al., 2010; Watling and Braga, 2015). At the local scale, even if the amphibian reaches its goal, the new patch may not have water resources available for the species reproduction, i.e. to perform its specific reproductive mode ("dry" fragments; Becker et al., 2010). The remaining habitat may present a history of human-made disturbances (e.g., intentional fire, logging), which can remove part of the habitat required for amphibians to survive (e.g., leaf-litter for direct-developing species, epiphytic bromeliads for bromeligenous species), reducing habitat suitability for most species (Pilliod et al., 2003; Gardner et al., 2007; Burivalova, Şekercioğlu, & Koh, 2014). Forest-specialist species may leave the patch if it does not have the necessary habitat to survive and reproduce, which may result in a higher mortality during migration through matrix (Becker et al., 2010). In the present study, matrix type explained close to 25% of the total variance in the respective model, and its effect size varied according to matrix type. There is variability among the capacity of these species to cross human-made environments, due to mobility and physiological tolerance (Gascon et al., 1999; Dixo and Metzger, 2010; Nowakowski et al., 2015; Watling and Braga, 2015). Generalist species are usually more adapted to open habitats and inhabit the remnant's edge, while forest-specialists avoid the environment outside the remnant usually preferring more central portions (Becker et al., 2010; Dixo and Metzger, 2010; Thompson et al., 2016; Pfeifer et al., 2017). This gradient of habitat tolerance could explain the divergent effect-sizes between matrix types.

In contrast, clear-cut and silviculture represents the most influential matrix types in our meta-analysis (Table 4). But they were also the least represented ($k < 3$). Clear-cut matrix was represented only by the Zimmerman and Bierregaard (1986) study, which came from the Biological Dynamics of Forest Fragments Project (INPE-WWF), a long term project in Brazil's Amazon forest specially designed to test the species-area relationship with forest patches. The study's high correlation ($r = 0.99$) was attributed to the availability of breeding habitats inside the larger remnants, where half of the frog species were habitat specialists and another half were less demanding about reproductive habitat. Silviculture was considered to have some conservation value, so long as they preserve part of the original forest or at least kept a

relatively hospitable environment for some species to explore and disperse through. However, this is not the case among types and taxa (e.g., Volpato et al., 2010; Kellner and Swihart, 2014). Silviculture had no distinct effect on the herpetofauna richness in comparison with natural habitats (Thompson et al., 2016). Despite our results showing a significant positive effect for clear-cut and silviculture matrices, more studies regarding their effects in a SAR context are needed to confirm these tendencies. Similarly, flooding matrix may not have been significant due to the low number of studies found ($k = 2$).

Agricultural landscapes proved to have a significant and homogeneous effect on the SAR ($I^2 = 26.9\%$), but with a lower effect size (0.24) in comparison to grazing and urban matrices (0.78 and 0.82, respectively). Watling and Donnelly (2006) also tested the influence of the surrounding matrices in SAR, and reached a similar result when considering multiple taxa (invertebrates, birds, herps and mammals, $k = 118$). In their study, agriculture matrix (crops plus grazing) had the lowest z slope ($= 0.18$). Agricultural matrices may provide low habitat heterogeneity for amphibians (e.g., Souza et al., 2014; Figueiredo et al., 2019). Furthermore, agrosystems are usually subject to periodic exposure to pesticides and fertilizers, which may cause deleterious effects such as deformations, decrease reproduction and increase mortality (Mann et al., 2009). Even protected areas are subject to these chemical substances, despite buffer zones (García-Muñoz et al., 2019; Sawatzky and Fahrig, 2019). Both agriculture and grazing matrices were previously associated with high species turn-over, with greater substitution of specialist species to generalist species when compared to natural habitats (Gascon et al., 1999; Thompson et al., 2016). Although, the abundance and richness of generalist amphibian species are greater in grazing than in agriculture systems (Thompson et al., 2016). Grazing systems consist predominantly of maintained grassy plants, but in contrast to crop monocultures system, allow the occurrence of scattered trees and ponds, acting as refugia and support for farm animals. Trees and ponds allow some degree of connectivity to less tolerant species across open environments (Ribeiro et al., 2011; Prevedello et al., 2018; Jeliaskov et al., 2019) as they provide cover and humidity refugia.

Urban matrices had a mean effect size similar to grazing matrices, but demonstrated a higher heterogeneity among studies in comparison to agricultural and grazing systems (Fig. 2, Table 4). The major characteristic of human occupation is the high human density, associated with physical and chemical alterations inside and around patches. Most studies determined a negative effect of urbanization on amphibian richness, abundance, occupancy and community structure (as reviewed by Hamer and McDonnell, 2008). Urbanization modifies the amphibians' breeding habitats and hydroperiod by the construction of dams, artificial channels or impermeable surfaces (Hamer and McDonnell, 2008). The habitat may also reduce its suitability to amphibian reproduction due to water and noise pollution (Bee and Swanson, 2007; Sievers et al., 2018; Sievers et al., 2019), which enable only highly tolerant species to utilize some of the urban breeding sites (Menin et al., 2019). Moreover, the high road density have a significant negative effect on amphibians, by limiting species mobility and reproduction (Fahrig et al., 1995; Mazerolle, 2004). Nevertheless, the occurrence of species in urban areas was especially related to the availability and heterogeneity of breeding habitats, sometimes more than patch size and shape (Hamer and McDonnell, 2008; Li et al., 2018). Both grazing and urban systems, despite their ecological differences, lead to the replacement of specialist species by highly tolerant generalist species in the context of habitat loss, which may explain their similar effect sizes. The high structure variability between urban patches may result in miscellaneous effect sizes regarding the SAR, and a higher heteroscedasticity between studies as consequence.

5. Conclusions

Since the implementation of Island Biogeography theory in conservation ecology (Diamond, 1975), the theory is one of the main topics

debated in this respective area. Yet, thinking in a global and general model focused only in patch size and isolation may be a "caricature of reality" (Laurance, 2008), because matrix type surrounding the patches also has a large amount of influence in the species richness and composition (Ricketts, 2001; Laurance, 2008, 2010; Dixo and Metzger, 2010). Our data integrate this fact, by suggesting that forest patch size have distinct degrees of influence in amphibian richness across multiple landscapes worldwide. One quarter of the between-studies variance can be attribute to the distinct matrix surroundings alone.

Protective laws and guidelines to create preserved areas may be strongly based on the "natural reserve design guidelines", suggested by Diamond (1975), but they usually ignore the landscape context (Fahrig, 2019). Species did not restrict themselves into the patches, and biodiversity threats do not respect the protected-area boundaries (Laurance et al., 2012; Pfeifer et al., 2017; Sawatzky and Fahrig, 2019). Creating and restoring protected areas is a challenge for decision-makers, particularly in developing and forested countries, because of conflicts with important social and economic issues (Bergallo et al., 2009; Bustamante et al., 2019). A possible alternative to this problem is the creation of laws for biodiversity-based land management (Foley et al., 2005; Kremen and Merenlender, 2018), where the land use surrounding protected areas may be managed as well.

Before any discussion regarding the divergent responses of fauna to patch size, we suggest an investigation on the local landscape first, followed by a discussion of how this could affect different local taxa. We are not proposing the adoption of one particular kind of matrix habitat for management purposes, but remarking that they affect the SAR and need to be considered in future landscape ecology studies. There is no doubt that patch size is an important driver for species occupancy and abundance, but it needs to be interpreted in a specific landscape context. If possible, patch size should be compared within a context of multiple landscapes, and not only in a patch-focused and local approach. The scale effects of patch size, habitat amount and especially fragmentation *per se*, were intensely discussed between Fahrig (2017); Fletcher et al. (2018); Fahrig et al. (2019); and Miller-Rushing et al. (2019).

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